EX NO: 5 Develop a customized ping command to test the server connectivity.

Aim:

To develop a customized ping command in python that checks the connectivity between a client and a server by sending ICMP echo request packets and displaying the response time and status.

Theory:

A ping command is a network diagnostic tool used to determine whether a host (server) is reachable over a network.

It works using ICMP (Internet control message protocol)

The customized ping program allows the user to

* Enter any hostname or IP address
* Send multiple ping requests
* Display response time
* Show packet loss
* Indicate whether the server is reachable.

Working procedure.

1. Install python and VS code.
2. Create a python file named ping.py
3. Import the required modules (subprocess and platform.
4. Accept the hostname or IP address from the user.
5. Detect the operating system.

```
Command Prompt
Microsoft Windows [Version 10.0.26200.8973]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Deepika>python --version
Python 3.14.0

C:\Users\Deepika>
```

ping.py

```
import subprocess
import platform
print("=" * 50)
print(" Customized Ping Utility")
print("=" * 50)
host = input("Enter Website or IP Address: ")
system = platform.system()
if system == "Windows":
    command = ["ping", "-n", "4", host]
else:
    command = ["ping", "-c", "4", host]
try:
    result = subprocess.run(command, capture_output=True, text=True)
    print("\nPing Results\n")
    print(result.stdout)
    if result.returncode == 0:
        print("Server Status : Reachable")
    else:
        print("Server Status : Unreachable")
except Exception as e:
    print("Error:", e)

```

```
PS C:\Users\Deepika\Downloads\CustomizedPing> python ping.py
==========================================
 Customized Ping Utility
==========================================
Enter Website or IP Address: google.com

Ping Results


Pinging google.com [142.251.221.174] with 32 bytes of data:
Reply from 142.251.221.174: bytes=32 time=5ms TTL=119
Reply from 142.251.221.174: bytes=32 time=5ms TTL=119
Reply from 142.251.221.174: bytes=32 time=7ms TTL=119
Reply from 142.251.221.174: bytes=32 time=6ms TTL=119

Ping statistics for 142.251.221.174:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 5ms, Maximum = 7ms, Average = 5ms

Server Status : Reachable
```

```
PS C:\Users\Deepika\Downloads\CustomizedPing> python ping.py
==========================================
 Customized Ping Utility
==========================================
Enter Website or IP Address: ajfbjegeufeugwegeg.com

Ping Results

Ping request could not find host ajfbjegeufeugwegeg.com. Please check the name and try again.

Server Status : Unreachable
```

